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LAYOUT METHOD AND MEMORY MACRO INCLUDING THROUGH-SILICON VIA

Abstract

A memory macro includes a first memory array, a second memory array, a word line driver between the first memory array and the second memory array, and a control circuit adjacent to and coupled to the word line driver, and one of the word line driver or the control circuit comprises a through-silicon via (TSV) surrounded by a dielectric layer within the corresponding word line driver or control circuit.

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Background/Summary

PRIORITY CLAIM [0001] The present application is a continuation of U.S. application Ser. No. 18/524,668, filed Nov. 30, 2023, now U.S. Pat. No. 12,308,303, issued May 20, 2025, which is a continuation of U.S. application Ser. No. 18/153,475, filed Jan. 12, 2023, now U.S. Pat. No. 11,854,943, issued Dec. 26, 2023, which is a continuation of U.S. application Ser. No. 17/209,878, filed Mar. 23, 2021, now U.S. Pat. No. 11,562,946, issued Jan. 24, 2023, which claims the priority of China Application No. 202110263207.8, filed Mar. 11, 2021, which are incorporated herein by reference in their entireties.

BACKGROUND

[0002] The ongoing trend in miniaturizing integrated circuits (ICs) has resulted in progressively smaller devices which consume less power, yet provide more functionality at higher speeds than earlier technologies. Such miniaturization has been achieved through design and manufacturing innovations tied to increasingly strict specifications.

[0003] IC packages are often used for applications in which power is distributed among one or more IC dies. In some cases, dies are stacked in three-dimensional (3D) arrangements in which power distribution relies on through-silicon vias (TSVs) in one or more of the stacked IC dies.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIGS. 1A and 1B are diagrams of a memory macro structure, in accordance with some embodiments.

[0006] FIG. 2 is a diagram of a memory macro structure, in accordance with some embodiments.

[0007] FIGS. 3A-3C are diagrams of portions of memory macro structures, in accordance with some embodiments.

[0008] FIG. 4 is a diagram of an IC package, in accordance with some embodiments.

[0009] FIG. 5 is a flowchart of a method of operating an IC package, in accordance with some embodiments.

[0010] FIG. 6 is a flowchart of a method of manufacturing a memory macro structure, in accordance with some embodiments.

[0011] FIG. 7 is a flowchart of a method of generating an IC layout diagram, in accordance with some embodiments.

[0012] FIGS. 8A-8C are IC layout diagrams, in accordance with some embodiments.

[0013] FIG. 9 is a block diagram of an IC layout diagram generation system, in accordance with some embodiments.

[0014] FIG. 10 is a block diagram of an IC manufacturing system, and an IC manufacturing flow associated therewith, in accordance with some embodiments.

DETAILED DESCRIPTION

[0015] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components, values, operations, materials, arrangements, or the like, are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. Other components, values, operations, materials, arrangements, or the like, are contemplated. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0016] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0017] In various embodiments, a memory macro structure includes a TSV extending through the memory macro and spanning front and back sides of a memory die that includes the memory macro. A power distribution structure of an IC package including the memory die positioned between a logic die and a substrate is thereby capable of including TSVs that extend both between and through memory macros. Compared to approaches that do not include a TSV extending through a memory macro, TSV density is increased such that resistance and power losses in the power distribution structure are decreased for a given memory macro size.

[0018] FIGS. 1A and 1B are diagrams of a memory macro structure **100**, in accordance with some embodiments. FIG. 1A depicts a plan view including X and Y directions, and FIG. 1B depicts a cross-sectional view along a plane A-A' that includes the X direction and a Z direction.

[0019] Memory macro structure **100** includes multiple instances of memory macros **100M** and TSVs **100T**; each instance of TSV **100T** spans a front side FS and a back side BS of an IC die **100D**, also referred to as memory die **100D** in some embodiments. In the embodiment depicted in FIGS. 1A and 1B, each instance of memory macro **100M** includes two instances of TSV **100T** extending through, and electrically isolated from, the instance of memory macro **100M**, as discussed below. Memory macro structure **100M** and IC die **100D** are usable as components of an IC package, e.g., an IC package **400** discussed below with respect to FIG. 4.

[0020] The numbers, locations, and relative sizes of memory macros **100M** and TSVs **100T** depicted in FIGS. 1A and 1B are non-limiting examples provided for the purpose of illustration. In various embodiments, memory macro structure **100** includes memory macros **100M** and TSVs **100T** having numbers, locations, and/or relative sizes other than those depicted in FIGS. 1A and 1B.

[0021] A memory macro, e.g., memory macro **100M**, is a memory circuit including at least one

array of memory cells configured to store data, and one or more circuits configured to control data input, output, and storage operations (details not depicted in FIGS. 1A and 1B). In some embodiments, memory cells of memory macro **100M** include static random-access memory (SRAM) cells. In various embodiments, SRAM cells include five-transistor (5T) SRAM cells, six-transistor (6T) SRAM cells, eight-transistor (8T) SRAM cells, nine-transistor (9T) SRAM cells, or SRAM cells having other numbers of transistors. In various embodiments memory cells of memory macro **100M** include dynamic random-access memory (DRAM) cells, read-only memory (ROM) cells, non-volatile memory (NVM) cells, or other memory cell types capable of storing data.

[0022] A TSV, e.g., TSV **100T**, is a conductive structure that spans front and back sides of an IC die, e.g., front side FS and back side BS of IC die **100D**, and is thereby configured to provide a low resistance path through the IC die. A TSV includes one or more conductive materials, e.g., copper, aluminum, tungsten, titanium, and/or other material(s) suitable for providing a low-resistance path between front and back sides of an IC die. By being configured to provide a low resistance path through the IC die, a TSV is capable of being included in a power distribution structure of an IC package, e.g., a power distribution structure **400PDS** of IC package **400** discussed below with respect to FIG. 4.

[0023] An instance of memory macro **100M** including one or more instances of TSV **100T** extending through the memory macro **100M** is also referred to as memory macro structure **100M**. In some embodiments, one or more instances of memory macro structure **100M** includes memory macro structure **200** discussed below with respect to FIGS. 2-3C.

[0024] In the embodiment depicted in FIGS. 1A and 1B, each instance of memory macro **100M** includes two instances of TSV **100T** extending through, and electrically isolated from, the instance of memory macro **100M**. In various embodiments, a given instance of memory macro **100M** includes zero, one, or greater than two instances of TSV **100T** extending through, and electrically isolated from, the instance of memory macro **100M**.

[0025] In the embodiment depicted in FIGS. 1A and 1B, memory macro structure **100**, also referred to as memory die structure **100** in some embodiments, includes each of memory macros **100M** and TSVs **100T** arranged in rows along the X direction. The rows of TSVs **100T** are positioned both within memory macros **100M** of the rows of memory macros **100M** and between adjacent rows of memory macros **100M**. TSVs **100T** are centered in memory macros **100M** in the Y direction such that TSVs **100T** have a pitch **P1** in the Y direction. A pitch **P2** in the Y direction of memory macros **100M** is thereby twice as large as pitch **P1**.

[0026] For a given size of memory macro **100M**, by including TSVs **100T** having pitch **P1** half as large as pitch **P2**, memory macro structure **100** includes TSVs **100T** having a greater density than densities in approaches that do not include TSVs extending through memory macros. In some embodiments, memory macro structure **100** includes TSVs **100T** otherwise arranged to include at least one TSV **100T** extending through, and electrically isolated from, at least one memory macro **100M** such that a density of TSVs **100T** is greater than densities in approaches that do not include TSVs extending through memory macros.

[0027] In various embodiments, memory macro structure **100** includes TSVs **100T** positioned other than centered in memory macros **100M** in the Y direction and/or positioned between adjacent columns of memory macros **100M** instead of and/or in addition to between adjacent rows of memory macros **100M**. In various embodiments, memory macro structure **100** includes subsets of memory macros **100M**, e.g., alternating rows and/or columns, in which a first subset includes one or more TSVs **100T** and a second subset is free from including one or more TSVs **100T**.

[0028] By including at least one TSV **100T** extending through, and electrically isolated from, at least one memory macro **100M** such that a density of TSVs **100T** is greater than densities in approaches that do not include TSVs extending through memory macros, IC die **100D** including the at least one memory macro **100M** is capable of being included in an IC package, e.g., IC package **400** discussed below with respect to FIG. 4, in which resistance and power losses in the

power distribution structure are decreased for a given memory macro size.

[0029] FIG. 2 is a diagram of memory macro structure **200**, in accordance with some embodiments. Memory macro structure **200** is usable as one or more instances of memory macro **100M** discussed above with respect to FIGS. 1A and 1B. FIG. 2 depicts a plan view of memory macro structure **200** including the X and Y directions discussed above with respect to FIGS. 1A and 1B. Each of FIGS. 3A-3C discussed below is a diagram of a portion of memory macro structure **200**, in accordance with some embodiments.

[0030] Memory macro structure **200** includes a global control circuit **200GCT**, global input/output (I/O) circuits **200GIO**, local control circuits **200LCT**, local I/O circuits **200LIO**, cell activation circuits **200WLD**, memory arrays **200A**, and TSVs **100T** discussed above with respect to FIGS. 1A and 1B.

[0031] Global control circuit **200GCT** is positioned between and electrically coupled to global I/O circuits **200GIO**, and electrically coupled to each instance of local control circuit **200LCT**. Each instance of local control circuit **200LCT** is positioned between and electrically coupled to two instances of local I/O circuit **200LIO**, and positioned between and electrically coupled to two instances of cell activation circuit **200WLD**, also referred to as word line driver **200WLD** in some embodiments. Each instance of local I/O circuit **200LIO** and each instance of cell activation circuit **200WLD** is positioned between and electrically coupled to two instances of memory array **200A**. In various embodiments, memory macro structure **200** includes combinations of one or more of address lines, bit lines, data lines, cell activation lines (also referred to as word lines in some embodiments), and/or signal lines (not shown in FIG. 2) whereby global control circuit **200GCT**, global I/O circuits **200GIO**, local control circuits **200LCT**, local I/O circuits **200LIO**, cell activation circuits **200WLD**, and memory arrays **200A** are electrically coupled to each other as discussed.

[0032] Memory arrays **200A** are arrays of memory cells configured to store data as discussed above with respect to FIGS. 1A and 1B. Each of global control circuit **200GCT**, global I/O circuits **200GIO**, local control circuits **200LCT**, local I/O circuits **200LIO**, and cell activation circuits **200WLD** is an IC configured to perform a subset of operations whereby data are input to, output from, and stored in corresponding instances of memory array **200A** responsive to various combinations of address, clock, control, and/or data signals (not shown in FIG. 2).

[0033] Global control circuit **200GCT** is configured to generate and receive one or more of the address, clock, control, and/or data signals configured to control top-level operation of memory macro structure **200**; each instance of global I/O circuit **200GIO** is configured to, responsive to one or more of the address, clock, control, and/or data signals, perform top-level I/O operations; each instance of local control circuit **200LCT** is configured to, responsive to one or more of the address, clock, control, and/or data signals, control operation of adjacent instances of local I/O circuit **200LIO** and cell activation circuit **200WLD**, thereby controlling diagonally adjacent instances of memory array **200A**; and each instance of local I/O circuit **200LIO** and cell activation circuit **200WLD** is configured to, responsive to one or more of the address, clock, control, and/or data signals, partially control operation of adjacent instances of memory array **200A**.

[0034] In the embodiment depicted in FIG. 2, memory macro structure **200** includes a total of two instances of local control circuit **200LCT**, each corresponding to four diagonally adjacent instances of memory array **200A**. In various embodiments, memory macro structure **200** includes a total of one or more than two instances of local control circuit **200LCT**, each corresponding to four diagonally adjacent instances of memory array **200A**. In some embodiments, memory macro structure **200** includes at least one instance of local control circuit **200LCT** corresponding to fewer or greater than four instances of memory array **200A**.

[0035] In the embodiment depicted in FIG. 2, a single instance of TSV **100T** extends through, and is electrically isolated from, global control circuit **200GCT**, each instance of local control circuit **200LCT**, and each instance of cell activation circuit **200WLD**. In some embodiments, more than one instance of TSV **100T** extends through, and is electrically isolated from, one or more of global

control circuit **200GCT**, each instance of local control circuit **200LCT**, and each instance of cell activation circuit **200WLD**. In some embodiments, one or more of global control circuit **200GCT**, each instance of local control circuit **200LCT**, and each instance of cell activation circuit **200WLD** is free from including a TSV **100T**.

[0036] In some embodiments, one or more instances of TSV **100T** extends through, and is electrically isolated from, global control circuit **200GCT** and each instance of local control circuit **200LCT**; and each instance of cell activation circuit **200WLD** is free from including a TSV **100T**. In some embodiments, one or more instances of TSV **100T** extends through, and is electrically isolated from, each instance of cell activation circuit **200WLD**; and each of global control circuit **200GCT** and each instance of local control circuit **200LCT** is free from including a TSV **100T**.

[0037] In various embodiments, one or more instances of TSV **100T** (not shown) extends through, and is electrically isolated from, each of one or more instances of global I/O circuit **200GIO**, local I/O circuit **200LIO**, and/or memory array **200A**.

[0038] In each of the embodiments depicted in FIGS. 3A-3C, an instance of cell activation circuit **200WLD** is adjacent to and electrically coupled to each of two instances of memory array **200A**, and an instance of local control circuit **200LCT** is adjacent to, e.g., in the positive or negative X direction, the instance of cell activation circuit **200WLD**. Cell activation circuit **200WLD** includes a portion **200WLDA** electrically coupled to the first instance of memory array **200A** and a portion **200WLDB** electrically coupled to the second instance of memory array **200A**.

[0039] In each of the embodiments depicted in FIGS. 3A and 3B, cell activation circuit **200WLD** includes an instance of a dummy area **200D**, and an instance of TSV **100T** extends through the instance of dummy area **200D** and is thereby electrically isolated from cell activation circuit **200WLD**. In some embodiments, control circuit **200LCT** includes an instance of dummy area **200D**, an additional instance of TSV **100T** extends through the instance of dummy area **200D** of control circuit **200LCT**, and the additional instance of TSV **100T** is electrically isolated from control circuit **200LCT**. In the embodiment depicted in FIG. 3C, control circuit **200LCT** includes an instance of dummy area **200D**, and an instance of TSV **100T** extends through the instance of dummy area **200D** and is thereby electrically isolated from control circuit **200LCT**, and cell activation circuit **200WLD** is free from including a TSV **100T**.

[0040] In each of the embodiments depicted in FIGS. 3A and 3B, local control circuit **200LCT** is coupled to portion **200WLDA** through a signal bus CTLBA and separately coupled to portion **200WLDB** through a signal bus CTLBB. Local control circuit **200LCT** is thereby configured to separately communicate a first set of signals CTLA to portion **200WLDA** through signal bus CTLBA and a second set of signals CTLB to portion **200WLDB** through signal bus CTLBB.

[0041] In some embodiments, each of portions **200WLDA** and **200WLDB** includes an address decoder and each of first set of signals CTLA and second set of signals CTLB includes one or more sets of pre-decode signals.

[0042] In the embodiment depicted in FIG. 3C, local control circuit **200LCT** is coupled to both of portions **200WLDA** and **200WLDB** through a single signal bus CTLB, and local control circuit **200LCT** is thereby configured to communicate a set of signals CTL to both of portions **200WLDA** and **200WLDB** through signal bus CTLB. In some embodiments, each of portions **200WLDA** and **200WLDB** includes an address decoder and set of signals CTL includes one or more sets of pre-decode signals.

[0043] In the embodiment depicted in FIG. 3A, an instance of dummy area **200D** extends across an entirety of cell activation circuit **200WLD** such that portions **200WLDA** and **200WLDB** are separated by the instance of dummy area **200D**. In the embodiment depicted in FIG. 3B, an instance of dummy area **200D** extends across a portion of cell activation circuit **200WLD** such that portions **200WLDA** and **200WLDB** share first and second borders (not labeled) separated by the instance of dummy area **200D**.

[0044] In some embodiments, an instance of dummy area **200D** extends across a portion of cell

activation circuit **200WLD** such that portions **200WLDA** and **200WLDB** share a single border (not labeled) adjacent to the instance of dummy area **200D**. In some embodiments, cell activation circuit **200WLD** includes one or more additional instances of dummy area **200D** (not shown) such that portions **200WLDA** and **200WLDB** share one or more borders adjacent to each instance of dummy area **200D**.

[0045] In the embodiments depicted in FIGS. **3A** and **3B** a single instance of TSV **100T** extends through the instance of dummy area **200D** in cell activation circuit **200WLD**. In various embodiments, two or more instances of TSV **100T** extend through the instance of dummy area **200D** in cell activation circuit **200WLD** or the instance of dummy area **200D** in cell activation circuit **200WLD** is free from including a TSV **100T**.

[0046] By the configuration discussed above, memory macro structure **200** is capable of including at least one TSV **100T** extending through, and electrically isolated from, the memory macro structure **200** such that an IC die including memory macro structure **200** is capable of realizing the benefits discussed above with respect to memory macro structure **100**.

[0047] FIG. **4** is a diagram of IC package **400**, in accordance with some embodiments. FIG. **4** depicts a cross-sectional view of IC package **400** including the X and Z directions discussed above with respect to FIGS. **1A** and **1B**. IC package **400** is a non-limiting example of an IC package including at least one instance of IC die **100D** in which one or more instances of TSV **100T** extend through one or more instances of memory macro **100M** as discussed above with respect to FIGS. **1A-3C**.

[0048] IC package **400** includes a logic die **400L**, a substrate **400S**, memory dies **100D0-100D3** positioned between logic die **400L** and substrate **400S**, and power distribution structure **400PDS**.

[0049] Each of memory dies **100D0-100D3** is an instance of IC die **100D** including one or more instances of TSV **100T** extending through one or more instances of memory macro **100M** (representative instances labeled), each discussed above with respect to FIGS. **1A-3C**. Power distribution structure **400PDS** includes bump structures **400B** and the instances of TSV **100T**, and is thereby configured to electrically couple logic die **400L** to substrate **400S**.

[0050] Memory die **100D0** is adjacent to logic die **400L**; memory dies **100D1A-100D1C** are aligned along the X direction, and each of memory dies **100D1A-100D1C** is adjacent to memory die **100D0**; memory die **100D2** is adjacent to each of memory dies **100D1A-100D1C**; and memory die **100D3** is adjacent to each of memory die **100D2** and substrate **400S**. An instance of TSV **100T** is positioned between memory dies **100D0**, **100D1A**, **100D1B**, and **100D2**, and an instance of TSV **100T** is positioned between memory dies **100D0**, **100D1B**, **100D1C**, and **100D2**.

[0051] Logic die **400L**, memory dies **100D0**, **100D1A**, **100D2**, and **100D3**, and substrate **400S** are aligned along the Z direction; logic die **400L**, memory dies **100D0**, **100D1B**, **100D2**, and **100D3**, and substrate **400S** are aligned along the Z direction; and logic die **400L**, memory dies **100D0**, **100D1C**, **100D2**, and **100D3**, and substrate **400S** are aligned along the Z direction.

[0052] Logic die **400L** is an IC chip including one or more IC devices, e.g., one or a combination of a logic circuit, a signal circuit, or application processor, a system on an IC (SoIC), a transmitter and/or receiver, an application-specific IC (ASIC), a large-scale integration (LSI) or very large-scale integration (VLSI) circuit, a voltage or current regulator, or the like.

[0053] Substrate **400S** is an IC chip or printed circuit board including conductive segments supported and electrically separated by a plurality of insulation layers and configured to receive one or more power supply voltages and a reference, e.g., ground, voltage, and distribute the one or more power supply voltages and reference voltage to one or more of bump structures **400B**.

[0054] Conductive segments include conductive lines, vias, contact pads, and/or under-bump metallization (UBM) structures including one or more conductive materials, e.g., a metal such as copper, aluminum, tungsten, or titanium, polysilicon, or another material capable of providing a low resistance path. Insulation layers include one or more dielectric materials, e.g., silicon dioxide, silicon nitride, or one or more high-k dielectric materials, molding compounds, or other materials

capable of electrically insulating adjacent conductive segments from each other.

[0055] Power distribution structure **400PDS**, also referred to as power distribution network **400PDS** in some embodiments, includes a plurality of conductive segments supported and electrically separated by a plurality of insulation layers and arranged in accordance with power delivery requirements, e.g., of logic die **400L**. In various embodiments, power distribution structure **400PDS** includes one or a combination of a TSV, e.g., TSV **100T**, a through-dielectric via (TDV), a power rail, a super power rail, a buried power rail, a contact pad, conductive segments arranged in a grid or mesh structure, or another arrangement suitable for distributing power to one or more IC devices.

[0056] The plurality of conductive segments are arranged so as to contact logic die **400L** and some or all of the instances of TSV **100T** included in some or all of memory dies **100D0-100D3** such that power distribution structure **400PDS** is configured to electrically couple logic die **400L** to substrate **400S** through the some or all of the instances of TSV **100T** and bump structures **400B**.

[0057] Bump structures **400B** are conductive structures that overlie and contact portions of substrate **400S**, thereby being configured to provide electrical connections between substrate **400S** and some or all of the instances of TSV **100T** included in memory die **100D3**. In some embodiments, bump structures **400B** include lead. In some embodiments, bump structures **400B** include lead-free materials such as tin, nickel, gold, silver, copper, or other materials suitable for providing electrical connections to external conductive elements.

[0058] In some embodiments, bump structures **400B** have substantially spherical shapes. In some embodiments, bump structures **400B** are controlled collapse chip connection (C4) bumps, ball grid array bumps, microbumps or the like.

[0059] In the non-limiting example depicted in FIG. 4, IC package **400** includes six instances of memory die **100D**, memory dies **100D0-100D3**, arranged in four rows positioned between logic die **400L** and substrate **400S** so as to electrically couple logic die **400L** to substrate **400S**. In various embodiments, IC package includes greater or fewer than six instances of memory die **100D** and/or includes instances of memory die **100D** otherwise arranged so as to electrically couple logic die **400L** to substrate **400S**. In some embodiments, IC package **400** includes a single instance of memory die **100D** positioned between logic die **400L** and substrate **400S** so as to electrically couple logic die **400L** to substrate **400S**.

[0060] In the non-limiting example depicted in FIG. 4, IC package **400** includes memory dies **100D0-100D3** oriented with front side FS further along the Z direction than back side BS (representative instance of memory die **100D3** labeled). In various embodiments, one or more of memory dies **100D0-100D3** has an opposite orientation of back side BS further along the Z direction than front side FS.

[0061] In the non-limiting example depicted in FIG. 4, IC package **400** includes single instances of each of logic die **400L** and substrate **400S**. In various embodiments, IC package **400** includes two or more instances of one or both of logic die **400L** or substrate **400S**, and the instances of memory die **100D** are arranged so as to electrically couple each instance of logic die **400L** to each instance of substrate **400S**.

[0062] In the non-limiting example depicted in FIG. 4, memory dies **100D0-100D3** include a number of instances of memory macro **100M** ranging from one to five. In various embodiments, one or more of memory dies **100D0-100D3** is free from including an instance of memory macro **100M** or includes a number of instances of memory macro **100M** greater than five.

[0063] In the non-limiting example depicted in FIG. 4, the instances of memory macro **100M** include a number of instances of TSV **100T** ranging from one to three. In various embodiments, one or more instances of memory macro **100M** includes a number of instances of TSV **100T** greater than three.

[0064] By the configuration discussed above, IC package **400** includes at least one instance of IC die **100D** in which one or more instances of TSV **100T** extend through one or more instances of

memory macro **100M** such that IC package **400** is capable of realizing the benefits discussed above with respect to memory macro **100**.

[0065] FIG. 5 is a flowchart of a method **500** of operating an IC package, in accordance with one or more embodiments. Method **500** is usable with an IC package, e.g., IC package **400** discussed above with respect to FIG. 4.

[0066] The sequence in which the operations of method **500** are depicted in FIG. 5 is for illustration only; the operations of method **500** are capable of being executed in sequences that differ from that depicted in FIG. 5. In some embodiments, operations in addition to those depicted in FIG. 5 are performed before, between, during, and/or after the operations depicted in FIG. 5. In some embodiments, the operations of method **500** are part of operating a circuit, e.g., a circuit that includes an IC package.

[0067] At operation **510**, in some embodiments, a power supply voltage is received at a first end of a TSV of a memory die. Receiving the power supply voltage at the first end of the TSV of the memory die includes the memory die being positioned in an IC package, and receiving the power supply voltage at the first end of the TSV of the memory die includes receiving the power supply voltage from a power distribution structure of the IC package. In some embodiments, receiving the power supply voltage at the first end of the TSV of the memory die includes receiving the power supply voltage from power distribution structure **400PDS** discussed above with respect to FIG. 4. In some embodiments, receiving the power supply voltage from the power distribution structure includes receiving the power supply voltage from a bump structure, e.g., a bump structure **400B** discussed above with respect to FIG. 4.

[0068] In some embodiments, the memory die is positioned between a logic die and a substrate configured to receive the power supply voltage, and receiving the power supply voltage at the first end of the TSV of the memory die includes receiving the power supply voltage at the substrate. In some embodiments, receiving the power supply voltage at the substrate includes receiving the power supply voltage at substrate **400S** of IC package **400** discussed above with respect to FIG. 4.

[0069] In some embodiments, receiving the power supply voltage at the first end of the TSV of the memory die includes receiving the power supply voltage at an end of TSV **100T** of memory die **100D** discussed above with respect to FIGS. 1A-4.

[0070] In some embodiments, the TSV is one TSV of a plurality of TSVs, and receiving the power supply voltage at the first end of the TSV includes receiving the power supply voltage at each TSV of the plurality of TSVs, e.g., multiple instances of TSV **100T** discussed above with respect to FIGS. 1A-4.

[0071] In some embodiments, the memory die is one memory die of a plurality of memory dies, and receiving the power supply voltage at the first end of the TSV of the memory die includes receiving the power supply voltage translated through one or more additional memory dies of the plurality of memory dies.

[0072] In some embodiments, receiving the power supply voltage at the first end of the TSV of the memory die includes receiving a reference, e.g., ground voltage.

[0073] At operation **520**, the power supply voltage is translated through a memory macro to a second end of the TSV. Translating the power supply voltage through the memory macro includes translating the power supply voltage electrically isolated from the memory macro. In some embodiments, translating the power supply voltage through the memory macro includes translating the power supply voltage through memory macro **100** discussed above with respect to FIGS. 1A-4.

[0074] In some embodiments, translating the power supply voltage through the memory macro includes translating the power supply voltage through a control circuit or cell activation circuit of the memory macro, e.g., global control circuit **200GCT**, local control circuit **200LCT**, or cell activation circuit **200WLD** discussed above with respect to FIGS. 2-3C.

[0075] In some embodiments, the TSV is one TSV of a plurality of TSVs included in the memory macro, and translating the power supply voltage through the memory macro includes translating the

power supply voltage to second ends of each TSV of the plurality of TSVs, e.g., multiple instances of TSV **100T** discussed above with respect to FIGS. **1A-4**.

[0076] In some embodiments, the memory macro is one memory macro of a plurality of memory macros, and translating the power supply voltage through the memory macro includes translating the power supply voltage through each memory macro of the plurality of memory macros, e.g., multiple instances of memory macro **100M** discussed above with respect to FIGS. **1A-4**.

[0077] In some embodiments, translating the power supply voltage through the memory macro includes translating the reference voltage through the memory macro.

[0078] At operation **530**, the power supply voltage is received at the logic die from the second end of the TSV. Receiving the power supply voltage at the logic die includes receiving the power supply voltage from the power distribution structure of the IC package, e.g., power distribution structure **400PDS** of IC package **400** discussed above with respect to FIG. **4**.

[0079] In some embodiments, the memory die is one memory die of a plurality of memory dies, and receiving the power supply voltage from the second end of the TSV includes receiving the power supply voltage translated through one or more additional memory dies of the plurality of memory dies.

[0080] In some embodiments, the TSV is one TSV of a plurality of TSVs, and receiving the power supply voltage from the second end of the TSV includes receiving the power supply voltage from the second end of each TSV of the plurality of TSVs, e.g., multiple instances of TSV **100T** discussed above with respect to FIGS. **1A-4**.

[0081] In some embodiments, receiving the power supply voltage from the second end of the TSV includes receiving the reference voltage from the second end of the TSV.

[0082] By executing some or all of the operations of method **500**, IC package operation includes translating a power supply voltage through a memory macro including a TSV, thereby obtaining the benefits discussed above with respect to memory macro structure **100**.

[0083] FIG. **6** is a flowchart of a method **600** of manufacturing a memory macro structure, in accordance with some embodiments. Method **600** is operable to form memory macro structure **100** discussed above with respect to FIGS. **1A** and **1B** and/or IC package **400** discussed above with respect to FIG. **4**. In some embodiments, operations of method **600** are a subset of operations of a method of forming an IC package, e.g., a 2.5D IC package, a 3D IC package, or an InFO package.

[0084] In some embodiments, the operations of method **600** are performed in the order depicted in FIG. **6**. In some embodiments, the operations of method **600** are performed in an order other than the order depicted in FIG. **6**. In some embodiments, one or more additional operations are performed before, during, and/or after the operations of method **600**. In some embodiments, performing some or all of the operations of method **600** includes performing one or more operations as discussed below with respect to IC manufacturing system **1000** and FIG. **10**.

[0085] At operation **610**, in some embodiments, a memory macro is built in a semiconductor wafer. In some embodiments, building the memory macro includes building memory macro **100M** die **100D** discussed above with respect to FIGS. **1A-4**.

[0086] Building the memory macro includes building a plurality of IC devices, e.g., transistors, logic gates, memory cells, interconnect structures, and/or other suitable devices, configured to operate as discussed above with respect to memory macro **100M**.

[0087] Building the memory macro includes performing a plurality of manufacturing operations, e.g., one or more of a lithography, diffusion, deposition, etching, planarizing, or other operation suitable for building the plurality of IC devices in the semiconductor wafer.

[0088] In some embodiments, building the memory macro includes building the memory macro including a dummy area, e.g., dummy area **200D** discussed above with respect to FIGS. **3A-3C**. Building the memory macro including the dummy area includes forming one or more dielectric layers, thereby configuring the dummy area to be electrically isolated from the memory macro, e.g., electrically isolated from global control circuit **200GCT**, local control circuit **200LCT**, or cell

activation circuit **200WLD** discussed above with respect to FIGS. 2-3C.

[0089] Forming one or more dielectric layers includes depositing one or more dielectric materials, e.g., silicon dioxide, silicon nitride, or one or more high-k dielectric materials, or other materials capable of electrically insulating adjacent conductive segments from each other. In various embodiments, depositing a dielectric material includes performing a physical vapor deposition (PVD) or chemical vapor deposition (CVD) process, a laser chemical vapor deposition (LCVD) process, an evaporation process, an electron beam evaporation (E-gun) process, or another suitable deposition process.

[0090] In some embodiments, operation **620** is performed prior to performing operation **610**, and building the memory macro includes forming the dummy area adjacent to one or more TSVs extending through the memory macro structure.

[0091] In some embodiments, building the memory macro in the semiconductor wafer includes building a plurality of memory macros, e.g., multiple instances of memory macro **100M** discussed above with respect to FIGS. 1A-4, in the semiconductor wafer.

[0092] In some embodiments, operation **610** is repeated such that building the memory macro in the semiconductor wafer includes building a plurality of memory macros in a corresponding plurality of semiconductor wafers, e.g., corresponding to memory dies **100D0-100D3** discussed above with respect to FIG. 4.

[0093] At operation **620**, in some embodiments, a TSV spanning front and back sides of the semiconductor wafer and extending through the memory macro is constructed. Constructing the TSV includes performing a plurality of manufacturing operations including depositing and patterning one or more photoresist layers, performing one or more etching processes, and performing one or more deposition processes whereby one or more conductive materials are configured to form a continuous, low resistance structure spanning the front and back sides of the semiconductor wafer.

[0094] In some embodiments, constructing the TSV spanning the front and back sides of the semiconductor wafer and extending through the memory macro includes constructing TSV **100T** spanning front side FS and back side BS of IC die **100D** and extending through memory macro **100M** discussed above with respect to FIGS. 1A-4.

[0095] In some embodiments, constructing the TSV extending through the memory macro includes constructing the TSV extending through a dummy area of the memory macro, e.g., dummy area **200D** discussed above with respect to FIGS. 3A-3C.

[0096] In some embodiments, operation **620** is performed prior to performing operation **610**, and constructing the TSV extending through the memory macro includes constructing the TSV extending through one or more dielectric layers of the semiconductor wafer, the one or more dielectric layers corresponding to the memory macro. In some embodiments, portions of each of operations are performed iteratively, whereby the TSV spanning front and back sides of the semiconductor wafer and extending through the memory macro is constructed.

[0097] In some embodiments, constructing the TSV spanning front and back sides of the semiconductor wafer and extending through the memory macro includes constructing a plurality of TSVs, e.g., multiple instances of TSV **100T** discussed above with respect to FIGS. 1A-4.

[0098] In some embodiments, operation **610** is repeated such that constructing the TSV spanning front and back sides of the semiconductor wafer and extending through the memory macro includes constructing a plurality of TSVs spanning front and back sides of a corresponding plurality of semiconductor wafers and extending through corresponding memory macros, e.g., multiple instances of TSV **100T** corresponding to memory dies **100D0-100D3** discussed above with respect to FIG. 4.

[0099] At operation **630**, in some embodiments, the TSV is connected to a power distribution structure of an IC package, thereby electrically connecting a logic die to a substrate. Connecting the TSV to the power distribution structure of the IC package includes performing one or more IC

package manufacturing operations whereby a portion or all of the semiconductor wafer including the TSV extending through the memory macro is connected to the power distribution structure of the IC package.

[0100] In various embodiments, the one or more IC package manufacturing operations include one or more of a die separation process, a molding injection or deposition, a bonding process, a metal deposition process, a solder process, an annealing process, or another process suitable for manufacturing an IC package.

[0101] In some embodiments, connecting the TSV to the power distribution structure of the IC package includes connecting an instance of TSV **100T** to power distribution structure **400PDS** discussed above with respect to FIG. 4.

[0102] In some embodiments, the TSV is one TSV of a plurality of TSVs, and connecting the TSV to the power distribution structure of the IC package includes connecting each TSV of the plurality of TSVs to the power distribution structure of the IC package, e.g., connecting multiple instances of TSV **100T** to power distribution structure **400PDS** discussed above with respect to FIG. 4.

[0103] In some embodiments, operation **630** is repeated such that TSVs of a plurality of semiconductor wafers are connected to the power distribution structure, e.g., connecting TSVs of one or more of IC dies **100D0-100D3** to power distribution structure **400PDS** discussed above with respect to FIG. 4.

[0104] The operations of method **600** are capable of being performed as a whole or as separate subsets of operations. For example, by performing some or all of operations **610** and **620**, a memory macro structure including a TSV extending through, and electrically isolated from, the memory macro is formed, thereby obtaining the benefits discussed above with respect to memory macro structure **100**. By performing some or all of operation **630** based on a memory macro structure formed in accordance with operations **610** and **620**, an IC package is formed in which the memory macro structure includes a TSV extending through, and electrically isolated from, the memory macro, thereby obtaining the benefits discussed above with respect to memory macro structure **100** and the with respect to IC package **400**.

[0105] FIG. 7 is a flowchart of a method **700** of generating an IC layout diagram, in accordance with some embodiments. In some embodiments, generating the IC layout diagram includes generating an IC layout diagram, e.g., an IC layout diagram **800A-800C** discussed below with respect to FIGS. 8A-8C, corresponding to a memory macro structure, e.g., memory macro structure **100** discussed above with respect to FIGS. 1A-4, manufactured based on the generated IC layout diagram.

[0106] In some embodiments, some or all of method **700** is executed by a processor of a computer. In some embodiments, some or all of method **700** is executed by a processor **902** of an IC layout diagram generation system **900**, discussed below with respect to FIG. 9.

[0107] Some or all of the operations of method **700** are capable of being performed as part of a design procedure performed in a design house, e.g., a design house **1020** discussed below with respect to FIG. 10.

[0108] In some embodiments, the operations of method **700** are performed in the order depicted in FIG. 7. In some embodiments, the operations of method **700** are performed simultaneously and/or in an order other than the order depicted in FIG. 7. In some embodiments, one or more operations are performed before, between, during, and/or after performing one or more operations of method **700**.

[0109] FIGS. 8A-8C depict non-limiting examples of respective IC layout diagrams **800A-800C** generated by executing one or more operations of method **700** as discussed below, in some embodiments. Each of IC layout diagrams **800A-800C** is simplified for the purpose of illustration. In various embodiments, one or more of IC layout diagrams **800A-800C** includes features in addition to those depicted in FIGS. 8A-8C, e.g., one or more transistor elements, vias, contacts, isolation structures, wells, conductive elements, or the like. In addition to respective IC layout

diagrams **800A-800C**, each of FIGS. **8A-8C** depicts the X and Y directions discussed above with respect to FIGS. **1A-4**.

[0110] At operation **710**, in some embodiments, a layout diagram of a memory macro is modified to include a dummy region. Modifying the layout diagram of the memory macro to include the dummy region includes the dummy region being usable in a manufacturing process as part of defining a dummy area in a memory macro manufactured based on the layout diagram of the memory macro. In some embodiments, modifying the layout diagram of the memory macro to include the dummy region includes modifying memory macro **100M** discussed above with respect to FIGS. **1A-4**.

[0111] In some embodiments, modifying the layout diagram of the memory macro to include the dummy region includes locating the dummy region in a control circuit region or a cell activation circuit region of the memory macro. In some embodiments, modifying the layout diagram of the memory macro to include the dummy region includes the dummy region being usable as part of defining an instance of dummy area **200D** discussed above with respect to FIGS. **2-3C**.

[0112] In some embodiments, modifying the layout diagram of the memory macro to include the dummy region includes modifying the layout diagram of the memory macro to include a plurality of dummy regions. In some embodiments, modifying the layout diagram of the memory macro to include the dummy region includes modifying IC layout diagram **800A** to include dummy regions **800DR** as depicted in FIGS. **8A-8C**. IC layout diagram **800A** corresponds to memory macro **100M** and each of dummy regions **800DR** corresponds to an instance of dummy area **200D** discussed above with respect to FIGS. **1A-4**.

[0113] In some embodiments, modifying the layout diagram of the memory macro includes receiving the memory macro from a storage device, e.g., a non-transitory, computer-readable storage medium **904** discussed below with respect to FIG. **9**. In some embodiments, modifying the layout diagram of the memory macro includes receiving the memory macro through a network interface, e.g., a network interface **912** discussed below with respect to FIG. **9**.

[0114] In some embodiments, modifying the layout diagram of the memory macro includes the memory macro being included in an intellectual property (IP) block. In some embodiments, modifying the layout diagram of the memory macro includes receiving the IP block, e.g., in the form of one or more electronic files transmitted over a network.

[0115] In some embodiments, modifying the layout diagram of the memory macro includes storing the memory macro in the storage device and/or transmitting the memory macro through the network interface.

[0116] At operation **720**, the layout diagram of the memory macro including the dummy region is received. In some embodiments, receiving the layout diagram of the memory macro includes receiving the layout diagram at an IC layout diagram generation system, e.g., IC layout diagram generation system **900** discussed below with respect to FIG. **9**.

[0117] In some embodiments, receiving the layout diagram of the memory macro includes receiving IC layout diagram **800A**.

[0118] In some embodiments, receiving the layout diagram of the memory macro includes receiving a plurality of layout diagrams of memory macros. In various embodiments, receiving the plurality of layout diagrams of memory macros includes the layout diagrams being the same or varying layout diagrams.

[0119] At operation **730**, the layout diagram of the memory macro is placed in a layout diagram of an IC die. In some embodiments, placing the layout diagram of the memory macro in the layout diagram of the IC die includes the layout diagram of the IC die corresponding to IC die **100D** discussed above with respect to FIGS. **1A-4**.

[0120] In some embodiments, the layout diagram is one layout diagram of a plurality of layout diagrams of memory macros, and placing the layout diagram of the memory macro in the layout diagram of the IC die includes placing the plurality of layout diagrams of memory macros in the

layout diagram of the IC die. In some embodiments, placing the plurality of layout diagrams of memory macros in the layout diagram of the IC die includes arranging the plurality of layout diagrams of memory macros in rows and/or columns.

[0121] In some embodiments, placing the layout diagram of the memory macro in the layout diagram of the IC die includes placing IC layout diagram **800A** in one of IC layout diagram **800B** depicted in FIG. **8B** or IC layout diagram **800C** depicted in FIG. **8C**.

[0122] At operation **740**, a plurality of TSV regions is arranged in the layout diagram of the IC die by placing a first TSV region of the plurality of TSV regions in the dummy region. In some embodiments, placing the first TSV region of the plurality of TSV regions in the dummy region includes placing the first TSV region corresponding to an instance of TSV **100T** discussed above with respect to FIGS. **1A-4**.

[0123] In some embodiments, the dummy region is a first dummy region of a plurality of dummy regions of the memory macro, and arranging the plurality of TSV regions in the layout diagram of the IC die includes placing a second TSV region of the plurality of TSV regions in a second dummy region of the plurality of dummy regions.

[0124] In some embodiments, the layout diagram of the memory macro is one layout diagram of a plurality of layout diagrams of memory macros, and arranging the plurality of TSV regions in the layout diagram of the IC die includes placing a TSV region of the plurality of TSV regions in each dummy region of a corresponding memory macro of the plurality of layout diagrams of memory macros.

[0125] In some embodiments, the layout diagram of the memory macro is one layout diagram of a plurality of layout diagrams of memory macros arranged in rows, and arranging the plurality of TSV regions in the layout diagram of the IC die includes placing a subset of the plurality of TSV regions between adjacent rows of the plurality of layout diagrams of memory macros.

[0126] In some embodiments, arranging the plurality of TSV regions in the layout diagram of the IC die includes arranging TSV regions **800TSV** in one of IC layout diagram **800B** depicted in FIG. **8B** or IC layout diagram **800C** depicted in FIG. **8C**. Each of TSV regions **800TSV** is a region in an IC layout diagram usable in the manufacturing process as part of defining a TSV, e.g., TSV **100T** discussed above with respect to FIGS. **1A-4**.

[0127] In the non-limiting example depicted in FIG. **8B**, arranging TSV regions **800TSV** includes placing an instance of TSV region **800TSV** in each instance of dummy region **800DR** of each instance of IC layout diagram **800A**. In the non-limiting example depicted in FIG. **8C**, arranging TSV regions **800TSV** includes placing an instance of TSV region **800TSV** in a single instance of dummy region **800DR** of each instance of IC layout diagram **800A**. In various embodiments, arranging TSV regions **800TSV** includes otherwise placing instances of TSV region **800TSV** in instances of dummy region **800DR** of instances of IC layout diagram **800A**, e.g., placing varying numbers of instances of TSV regions in instances of dummy region **800DR** for a given instance of IC layout diagram **800A**.

[0128] In some embodiments, arranging the plurality of TSV regions in the layout diagram of the IC die is based on one or more design criteria of a logic die, e.g., logic die **400L** discussed above with respect to FIG. **4**. In some embodiments, the one or more design criteria include power supply voltage drops based on resistance values of a plurality of TSVs corresponding to the plurality of TSV regions.

[0129] At operation **750**, in some embodiments, the IC layout diagram is generated and stored in a storage device. Generating the IC layout diagram is performed by a processor, e.g., processor **902** of IC layout diagram generation system **900** discussed below with respect to FIG. **9**.

[0130] In some embodiments, generating the IC layout diagram includes positioning one or more features (not shown), e.g., a contact, via, or conductive region, corresponding to one or more IC structures manufactured based on the one or more features and configured to provide electrical connections to one or more memory macros corresponding to the memory macro including the

dummy region.

[0131] In various embodiments, storing the IC layout diagram in the storage device includes storing the IC layout diagram in a non-volatile, computer-readable memory, e.g., a database, and/or includes storing the IC layout diagram over a network. In various embodiments, storing the IC layout diagram in the storage device includes storing the IC layout diagram in non-volatile, computer-readable storage medium **904** and/or over network **914** of IC layout diagram generation system **900**, discussed below with respect to FIG. **9**.

[0132] In various embodiments, generating and storing the IC layout diagram includes generating and storing one or more of IC layout diagrams **800A-800C**.

[0133] At operation **760**, in some embodiments, at least one of one or more semiconductor masks, or at least one component in a layer of a semiconductor IC is fabricated based on the IC layout diagram. Fabricating one or more semiconductor masks or at least one component in a layer of a semiconductor IC is discussed below with respect to IC manufacturing system **1000** and FIG. **10**.

[0134] In various embodiments, fabricating one or more semiconductor masks or at least one component in the layer of the semiconductor IC is based on one or more of IC layout diagrams **800A-800C**.

[0135] At operation **770**, in some embodiments, one or more manufacturing operations are performed based on the IC layout diagram. In some embodiments, performing one or more manufacturing operations includes performing one or more lithographic exposures based on the IC layout diagram. Performing one or more manufacturing operations, e.g., one or more lithographic exposures, based on the IC layout diagram is discussed below with respect to FIG. **10**.

[0136] In various embodiments, performing one or more manufacturing operations is based on one or more of IC layout diagrams **800A-800C**.

[0137] By executing some or all of the operations of method **700**, an IC layout diagram, e.g., IC layout diagram **800A-800C**, is generated corresponding to a memory macro structure in which a TSV extends through the memory macro structure, thereby realizing the benefits discussed above with respect to memory macro structure **100**. Further, by placing a layout diagram of a memory macro and separately arranging a plurality of TSV regions, design flexibility is improved over approaches in which arranging a plurality of TSV regions is not separate from placing a layout diagram of a memory macro.

[0138] FIG. **9** is a block diagram of IC layout diagram generation system **900**, in accordance with some embodiments. Methods described herein of designing IC layout diagrams in accordance with one or more embodiments are implementable, for example, using IC layout diagram generation system **900**, in accordance with some embodiments.

[0139] In some embodiments, IC layout diagram generation system **900** is a general purpose computing device including a hardware processor **902** and non-transitory, computer-readable storage medium **904**. Computer-readable storage medium **904**, amongst other things, is encoded with, i.e., stores, computer program code **906**, i.e., a set of executable instructions. Execution of instructions **906** by hardware processor **902** represents (at least in part) an EDA tool which implements a portion or all of a method, e.g., method **700** of generating an IC layout diagram described above (hereinafter, the noted processes and/or methods).

[0140] Processor **902** is electrically coupled to computer-readable storage medium **904** via a bus **908**. Processor **902** is also electrically coupled to an I/O interface **910** by bus **908**. Network interface **912** is also electrically connected to processor **902** via bus **908**. Network interface **912** is connected to a network **914**, so that processor **902** and computer-readable storage medium **904** are capable of connecting to external elements via network **914**. Processor **902** is configured to execute computer program code **906** encoded in computer-readable storage medium **904** in order to cause IC layout diagram generation system **900** to be usable for performing a portion or all of the noted processes and/or methods. In one or more embodiments, processor **902** is a central processing unit (CPU), a multi-processor, a distributed processing system, an application specific integrated circuit

(ASIC), and/or a suitable processing unit.

[0141] In one or more embodiments, computer-readable storage medium **904** is an electronic, magnetic, optical, electromagnetic, infrared, and/or a semiconductor system (or apparatus or device). For example, computer-readable storage medium **904** includes a semiconductor or solid-state memory, a magnetic tape, a removable computer diskette, a random access memory (RAM), a read-only memory (ROM), a rigid magnetic disk, and/or an optical disk. In one or more embodiments using optical disks, computer-readable storage medium **904** includes a compact disk-read only memory (CD-ROM), a compact disk-read/write (CD-R/W), and/or a digital video disc (DVD).

[0142] In one or more embodiments, computer-readable storage medium **904** stores computer program code **906** configured to cause IC layout diagram generation system **900** (where such execution represents (at least in part) the EDA tool) to be usable for performing a portion or all of the noted processes and/or methods. In one or more embodiments, computer-readable storage medium **904** also stores information which facilitates performing a portion or all of the noted processes and/or methods. In one or more embodiments, computer-readable storage medium **904** stores IC die library **907** of IC dies including IC layout diagram **800A** and/or **800B** discussed above with respect to FIGS. **8A-8C**.

[0143] IC layout diagram generation system **900** includes I/O interface **910**. I/O interface **910** is coupled to external circuitry. In one or more embodiments, I/O interface **910** includes a keyboard, keypad, mouse, trackball, trackpad, touchscreen, and/or cursor direction keys for communicating information and commands to processor **902**.

[0144] IC layout diagram generation system **900** also includes network interface **912** coupled to processor **902**. Network interface **912** allows IC layout diagram generation system **900** to communicate with network **914**, to which one or more other computer systems are connected. Network interface **912** includes wireless network interfaces such as BLUETOOTH, WIFI, WIMAX, GPRS, or WCDMA; or wired network interfaces such as ETHERNET, USB, or IEEE-1364. In one or more embodiments, a portion or all of noted processes and/or methods, is implemented in two or more IC layout diagram generation systems **900**.

[0145] IC layout diagram generation system **900** is configured to receive information through I/O interface **910**. The information received through I/O interface **910** includes one or more of instructions, data, design rules, libraries of standard cells, and/or other parameters for processing by processor **902**. The information is transferred to processor **902** via bus **908**. IC layout diagram generation system **900** is configured to receive information related to a UI through I/O interface **910**. The information is stored in computer-readable storage medium **904** as user interface (UI) **942**.

[0146] In some embodiments, a portion or all of the noted processes and/or methods is implemented as a standalone software application for execution by a processor. In some embodiments, a portion or all of the noted processes and/or methods is implemented as a software application that is a part of an additional software application. In some embodiments, a portion or all of the noted processes and/or methods is implemented as a plug-in to a software application. In some embodiments, at least one of the noted processes and/or methods is implemented as a software application that is a portion of an EDA tool. In some embodiments, a portion or all of the noted processes and/or methods is implemented as a software application that is used by IC layout diagram generation system **900**. In some embodiments, a layout diagram which includes standard cells is generated using a tool such as VIRTUOSO® available from CADENCE DESIGN SYSTEMS, Inc., or another suitable layout generating tool.

[0147] In some embodiments, the processes are realized as functions of a program stored in a non-transitory computer readable recording medium. Examples of a non-transitory computer readable recording medium include, but are not limited to, external/removable and/or internal/built-in storage or memory unit, e.g., one or more of an optical disk, such as a DVD, a magnetic disk, such

as a hard disk, a semiconductor memory, such as a ROM, a RAM, a memory card, and the like.

[0148] FIG. **10** is a block diagram of IC manufacturing system **1000**, and an IC manufacturing flow associated therewith, in accordance with some embodiments. In some embodiments, based on an IC layout diagram, at least one of (A) one or more semiconductor masks or (B) at least one component in a layer of a semiconductor integrated circuit is fabricated using IC manufacturing system **1000**.

[0149] In FIG. **10**, IC manufacturing system **1000** includes entities, such as a design house **1020**, a mask house **1030**, and an IC manufacturer/fabricator (“fab”) **1050**, that interact with one another in the design, development, and manufacturing cycles and/or services related to manufacturing an IC device **1060**. The entities in IC manufacturing system **1000** are connected by a communications network. In some embodiments, the communications network is a single network. In some embodiments, the communications network is a variety of different networks, such as an intranet and the Internet. The communications network includes wired and/or wireless communication channels. Each entity interacts with one or more of the other entities and provides services to and/or receives services from one or more of the other entities. In some embodiments, two or more of design house **1020**, mask house **1030**, and IC fab **1050** is owned by a single larger company. In some embodiments, two or more of design house **1020**, mask house **1030**, and IC fab **1050** coexist in a common facility and use common resources.

[0150] Design house (or design team) **1020** generates an IC design layout diagram **1022**. IC design layout diagram **1022** includes various geometrical patterns, e.g., an IC layout diagram discussed above. The geometrical patterns correspond to patterns of metal, oxide, or semiconductor layers that make up the various components of IC device **1060** to be fabricated. The various layers combine to form various IC features. For example, a portion of IC design layout diagram **1022** includes various IC features, such as an active region, gate electrode, source and drain, metal lines or vias of an interlayer interconnection, and openings for bonding pads, to be formed in a semiconductor substrate (such as a silicon wafer) and various material layers disposed on the semiconductor substrate. Design house **1020** implements a proper design procedure to form IC design layout diagram **1022**. The design procedure includes one or more of logic design, physical design or place and route. IC design layout diagram **1022** is presented in one or more data files having information of the geometrical patterns. For example, IC design layout diagram **1022** can be expressed in a GDSII file format or DFII file format.

[0151] Mask house **1030** includes data preparation **1032** and mask fabrication **1044**. Mask house **1030** uses IC design layout diagram **1022** to manufacture one or more masks **1045** to be used for fabricating the various layers of IC device **1060** according to IC design layout diagram **1022**. Mask house **1030** performs mask data preparation **1032**, where IC design layout diagram **1022** is translated into a representative data file (“RDF”). Mask data preparation **1032** provides the RDF to mask fabrication **1044**. Mask fabrication **1044** includes a mask writer. A mask writer converts the RDF to an image on a substrate, such as a mask (reticle) **1045** or a semiconductor wafer **1053**. The design layout diagram **1022** is manipulated by mask data preparation **1032** to comply with particular characteristics of the mask writer and/or requirements of IC fab **1050**. In FIG. **10**, mask data preparation **1032** and mask fabrication **1044** are illustrated as separate elements. In some embodiments, mask data preparation **1032** and mask fabrication **1044** can be collectively referred to as mask data preparation.

[0152] In some embodiments, mask data preparation **1032** includes optical proximity correction (OPC) which uses lithography enhancement techniques to compensate for image errors, such as those that can arise from diffraction, interference, other process effects and the like. OPC adjusts IC design layout diagram **1022**. In some embodiments, mask data preparation **1032** includes further resolution enhancement techniques (RET), such as off-axis illumination, sub-resolution assist features, phase-shifting masks, other suitable techniques, and the like or combinations thereof. In some embodiments, inverse lithography technology (ILT) is also used, which treats OPC as an

inverse imaging problem.

[0153] In some embodiments, mask data preparation **1032** includes a mask rule checker (MRC) that checks the IC design layout diagram **1022** that has undergone processes in OPC with a set of mask creation rules which contain certain geometric and/or connectivity restrictions to ensure sufficient margins, to account for variability in semiconductor manufacturing processes, and the like. In some embodiments, the MRC modifies the IC design layout diagram **1022** to compensate for limitations during mask fabrication **1044**, which may undo part of the modifications performed by OPC in order to meet mask creation rules.

[0154] In some embodiments, mask data preparation **1032** includes lithography process checking (LPC) that simulates processing that will be implemented by IC fab **1050** to fabricate IC device **1060**. LPC simulates this processing based on IC design layout diagram **1022** to create a simulated manufactured device, such as IC device **1060**. The processing parameters in LPC simulation can include parameters associated with various processes of the IC manufacturing cycle, parameters associated with tools used for manufacturing the IC, and/or other aspects of the manufacturing process. LPC takes into account various factors, such as aerial image contrast, depth of focus (“DOF”), mask error enhancement factor (“MEEF”), other suitable factors, and the like or combinations thereof. In some embodiments, after a simulated manufactured device has been created by LPC, if the simulated device is not close enough in shape to satisfy design rules, OPC and/or MRC are repeated to further refine IC design layout diagram **1022**.

[0155] It should be understood that the above description of mask data preparation **1032** has been simplified for the purposes of clarity. In some embodiments, data preparation **1032** includes additional features such as a logic operation (LOP) to modify the IC design layout diagram **1022** according to manufacturing rules. Additionally, the processes applied to IC design layout diagram **1022** during data preparation **1032** may be executed in a variety of different orders.

[0156] After mask data preparation **1032** and during mask fabrication **1044**, a mask **1045** or a group of masks **1045** are fabricated based on the modified IC design layout diagram **1022**. In some embodiments, mask fabrication **1044** includes performing one or more lithographic exposures based on IC design layout diagram **1022**. In some embodiments, an electron-beam (e-beam) or a mechanism of multiple e-beams is used to form a pattern on a mask (photomask or reticle) **1045** based on the modified IC design layout diagram **1022**. Mask **1045** can be formed in various technologies. In some embodiments, mask **1045** is formed using binary technology. In some embodiments, a mask pattern includes opaque regions and transparent regions. A radiation beam, such as an ultraviolet (UV) or EUV beam, used to expose the image sensitive material layer (e.g., photoresist) which has been coated on a wafer, is blocked by the opaque region and transmits through the transparent regions. In one example, a binary mask version of mask **1045** includes a transparent substrate (e.g., fused quartz) and an opaque material (e.g., chromium) coated in the opaque regions of the binary mask. In another example, mask **1045** is formed using a phase shift technology. In a phase shift mask (PSM) version of mask **1045**, various features in the pattern formed on the phase shift mask are configured to have proper phase difference to enhance the resolution and imaging quality. In various examples, the phase shift mask can be attenuated PSM or alternating PSM. The mask(s) generated by mask fabrication **1044** is used in a variety of processes. For example, such a mask(s) is used in an ion implantation process to form various doped regions in semiconductor wafer **1053**, in an etching process to form various etching regions in semiconductor wafer **1053**, and/or in other suitable processes.

[0157] IC fab **1050** is an IC fabrication business that includes one or more manufacturing facilities for the fabrication of a variety of different IC products. In some embodiments, IC fab **1050** is a semiconductor foundry. For example, there may be a manufacturing facility for the front end fabrication of a plurality of IC products (front-end-of-line (FEOL) fabrication), while a second manufacturing facility may provide the back end fabrication for the interconnection and packaging of the IC products (back-end-of-line (BEOL) fabrication), and a third manufacturing facility may

provide other services for the foundry business.

[0158] IC fab **1050** includes wafer fabrication tools **1052** configured to execute various manufacturing operations on semiconductor wafer **1053** such that IC device **1060** is fabricated in accordance with the mask(s), e.g., mask **1045**. In various embodiments, wafer fabrication tools **1052** include one or more of a wafer stepper, an ion implanter, a photoresist coater, a process chamber, e.g., a CVD chamber or LPCVD furnace, a CMP system, a plasma etch system, a wafer cleaning system, or other manufacturing equipment capable of performing one or more suitable manufacturing processes as discussed herein.

[0159] IC fab **1050** uses mask(s) **1045** fabricated by mask house **1030** to fabricate IC device **1060**. Thus, IC fab **1050** at least indirectly uses IC design layout diagram **1022** to fabricate IC device **1060**. In some embodiments, semiconductor wafer **1053** is fabricated by IC fab **1050** using mask(s) **1045** to form IC device **1060**. In some embodiments, the IC fabrication includes performing one or more lithographic exposures based at least indirectly on IC design layout diagram **1022**.

Semiconductor wafer **1053** includes a silicon substrate or other proper substrate having material layers formed thereon. Semiconductor wafer **1053** further includes one or more of various doped regions, dielectric features, multilevel interconnects, and the like (formed at subsequent manufacturing steps).

[0160] Details regarding an IC manufacturing system (e.g., system **1000** of FIG. **10**), and an IC manufacturing flow associated therewith are found, e.g., in U.S. Pat. No. 9,256,709, granted Feb. 9, 2016, U.S. Pre-Grant Publication No. 20150278429, published Oct. 1, 2015, U.S. Pre-Grant Publication No. 20140040838, published Feb. 6, 2014, and U.S. Pat. No. 7,260,442, granted Aug. 21, 2007, the entireties of each of which are hereby incorporated by reference.

[0161] In some embodiments, a memory macro includes a first memory array, a second memory array, a word line driver between the first memory array and the second memory array, and a control circuit adjacent to and coupled to the word line driver, wherein one of the word line driver or the control circuit comprises a TSV surrounded by a dielectric layer within the corresponding word line driver or control circuit. In some embodiments, the TSV surrounded by the dielectric layer is a first TSV surrounded by a first dielectric layer, and the other of the word line driver or the control circuit includes a second TSV surrounded by a second dielectric layer within the corresponding other of the word line driver or the control circuit. In some embodiments, the control circuit is a local control circuit, the TSV surrounded by the dielectric layer is a first TSV surrounded by a first dielectric layer, the memory macro includes a global control circuit aligned with the local control circuit and the word line driver, and the global control circuit includes a second TSV surrounded by a second dielectric layer within the global control circuit. In some embodiments, the word line driver is a first word line driver, the TSV surrounded by the dielectric layer is a first TSV surrounded by a first dielectric layer within the first word line driver, the memory macro includes a third memory array, fourth memory array, and a second word line driver between the third memory array and the fourth memory array and adjacent to and coupled to the control circuit, and the second word line driver includes a second TSV surrounded by a second dielectric layer within the second word line driver. In some embodiments, the control circuit includes a third TSV surrounded by a third dielectric layer within the control circuit. In some embodiments, the word line driver includes a first portion adjacent to and coupled to the first memory array and a second portion adjacent to and coupled to the second memory array, and the TSV surrounded by the dielectric layer between the first portion and the second portion. In some embodiments, each of the first memory array and the second memory array includes a plurality of SRAM cells.

[0162] In some embodiments, a method of manufacturing an IC device includes receiving an IC layout diagram of a memory macro, wherein the IC layout diagram of the memory macro includes a first memory array, a second memory array, a word line driver between the first memory array and the second memory array, a control circuit adjacent to the word line driver, and a dummy

region within one of the word line driver or the control circuit, placing a TSV region within the dummy region, and storing the IC layout diagram comprising the dummy region and the TSV region in a storage device. In some embodiments, the dummy region is a first dummy region in the IC layout diagram of the memory macro within the one of the word line driver or the control circuit, the IC layout diagram of the memory macro includes a second dummy region within the other of the word line driver or the control circuit, and placing the TSV region within the dummy region includes placing the TSV region being a first TSV in the first dummy region and placing a second TSV region in the second dummy region. In some embodiments, the control circuit is a local control circuit, the dummy region is a first dummy region in the IC layout diagram of the memory macro within the one of the word line driver or the control circuit, the IC layout diagram of the memory macro includes a global control circuit aligned with the local control circuit and the word line driver and a second dummy region within the global control circuit, and placing the TSV region within the dummy region includes placing the TSV region being a first TSV in the first dummy region and placing a second TSV region in the second dummy region. In some embodiments, the word line driver is a first word line driver, the dummy region is a first dummy region in the IC layout diagram of the memory macro within the first word line driver, the IC layout diagram of the memory macro includes a third memory array, a fourth memory array, and a second word line driver between the third memory array and the fourth memory array and adjacent to the control circuit, and a second dummy region within the second word line driver, and placing the TSV region within the dummy region includes placing the TSV region being a first TSV in the first dummy region and placing a second TSV region in the second dummy region. In some embodiments, the word line driver includes a first circuit portion adjacent to the first memory array, a second circuit portion adjacent to the second memory array, and the dummy region between the first portion and the second portion. In some embodiments, each of the first memory array and the second memory array includes an array of SRAM cells. In some embodiments, the method includes generating the IC layout diagram of the memory macro by adding the dummy region to a previously stored memory macro IC layout diagram.

[0163] In some embodiments, a method of manufacturing an IC device includes receiving a plurality of memory macro layout diagrams, wherein each memory macro layout diagram of the plurality of memory macro layout diagrams includes a first memory array, a second memory array, a word line driver between the first memory array and the second memory array, a control circuit adjacent to the word line driver, and a dummy region within one of the word line driver or the control circuit, arranging the plurality of memory macro layout diagrams in rows and columns in an IC layout diagram, arranging a plurality of TSV regions in the IC layout diagram by placing a TSV region of the plurality of TSV regions within the dummy region of each memory macro layout diagram of the plurality of memory macro layout diagrams, and storing the IC layout diagram comprising the plurality of memory macro layout diagrams and the plurality of TSV regions in a storage device. In some embodiments, arranging the plurality of TSV regions in the IC layout diagram includes placing additional TSV regions of the plurality of TSV regions between the rows of memory macro layout diagrams of the plurality of memory macro layout diagrams. In some embodiments, arranging the plurality of memory macro layout diagrams in rows in the IC layout diagram includes arranging the rows having a first pitch, and placing the TSV region of the plurality of TSV regions within the dummy region of each memory macro layout diagram of the plurality of memory macro layout diagrams and the placing the additional TSV regions of the plurality of TSV regions between the rows of memory macro layout diagrams of the plurality of memory macro layout diagrams include arranging the plurality of TSV regions having a second pitch half as large as the first pitch. In some embodiments, the arranging the plurality of TSV regions in the IC layout diagram includes arranging the plurality of TSV regions based on a power distribution structure of an IC package. In some embodiments, arranging the plurality of TSV regions in the IC layout diagram includes arranging the plurality of TSV regions based on one or

more design criteria of a logic die separate from the IC layout diagram. In some embodiments, each of the first memory array and the second memory array of each memory macro layout diagram of the plurality of memory macro layout diagrams comprises an array of SRAM cells.

[0164] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A memory macro comprising: a first memory array; a second memory array; a word line driver between the first memory array and the second memory array; and a control circuit adjacent to and coupled to the word line driver, wherein one of the word line driver or the control circuit comprises a through-silicon via (TSV) surrounded by a dielectric layer within the corresponding word line driver or control circuit.
2. The memory macro of claim 1, wherein the TSV surrounded by the dielectric layer is a first TSV surrounded by a first dielectric layer, and the other of the word line driver or the control circuit comprises a second TSV surrounded by a second dielectric layer within the corresponding other of the word line driver or the control circuit.
3. The memory macro of claim 1, wherein the control circuit is a local control circuit, the TSV surrounded by the dielectric layer is a first TSV surrounded by a first dielectric layer, the memory macro further comprises a global control circuit aligned with the local control circuit and the word line driver, and the global control circuit comprises a second TSV surrounded by a second dielectric layer within the global control circuit.
4. The memory macro of claim 1, wherein the word line driver is a first word line driver, the TSV surrounded by the dielectric layer is a first TSV surrounded by a first dielectric layer within the first word line driver, the memory macro further comprises: a third memory array; a fourth memory array; and a second word line driver between the third memory array and the fourth memory array and adjacent to and coupled to the control circuit, and the second word line driver comprises a second TSV surrounded by a second dielectric layer within the second word line driver.
5. The memory macro of claim 4, wherein the control circuit comprises a third TSV surrounded by a third dielectric layer within the control circuit.
6. The memory macro of claim 1, wherein the word line driver comprises: a first portion adjacent to and coupled to the first memory array and a second portion adjacent to and coupled to the second memory array; and the TSV surrounded by the dielectric layer between the first portion and the second portion.
7. The memory macro of claim 1, wherein each of the first memory array and the second memory array comprises a plurality of static random-access memory (SRAM) cells.
8. A method of manufacturing an integrated circuit (IC) device, the method comprising: receiving an IC layout diagram of a memory macro, wherein the IC layout diagram of the memory macro comprises: a first memory array; a second memory array; a word line driver between the first memory array and the second memory array; a control circuit adjacent to the word line driver; and a dummy region within one of the word line driver or the control circuit; placing a through-silicon via (TSV) region within the dummy region; and storing the IC layout diagram comprising the dummy region and the TSV region in a storage device.
9. The method of claim 8, wherein the dummy region is a first dummy region in the IC layout

diagram of the memory macro within the one of the word line driver or the control circuit, the IC layout diagram of the memory macro comprises a second dummy region within the other of the word line driver or the control circuit, and the placing the TSV region within the dummy region comprises placing the TSV region being a first TSV in the first dummy region and placing a second TSV region in the second dummy region.

10. The method of claim 8, wherein the control circuit is a local control circuit, the dummy region is a first dummy region in the IC layout diagram of the memory macro within the one of the word line driver or the control circuit, the IC layout diagram of the memory macro further comprises: a global control circuit aligned with the local control circuit and the word line driver; and a second dummy region within the global control circuit, and the placing the TSV region within the dummy region comprises placing the TSV region being a first TSV in the first dummy region and placing a second TSV region in the second dummy region.

11. The method of claim 8, wherein the word line driver is a first word line driver, the dummy region is a first dummy region in the IC layout diagram of the memory macro within the first word line driver, the IC layout diagram of the memory macro further comprises: a third memory array; a fourth memory array; and a second word line driver between the third memory array and the fourth memory array and adjacent to the control circuit; and a second dummy region within the second word line driver, and the placing the TSV region within the dummy region comprises placing the TSV region being a first TSV in the first dummy region and placing a second TSV region in the second dummy region.

12. The method of claim 8, wherein the word line driver comprises: a first circuit portion adjacent to the first memory array; a second circuit portion adjacent to the second memory array; and the dummy region between the first portion and the second portion.

13. The method of claim 8, wherein each of the first memory array and the second memory array comprises an array of static random-access memory (SRAM) cells.

14. The method of claim 8, further comprising: generating the IC layout diagram of the memory macro by adding the dummy region to a previously stored memory macro IC layout diagram.

15. A method of manufacturing an integrated circuit (IC) device, the method comprising: receiving a plurality of memory macro layout diagrams, wherein each memory macro layout diagram of the plurality of memory macro layout diagrams comprises: a first memory array; a second memory array; a word line driver between the first memory array and the second memory array; a control circuit adjacent to the word line driver; and a dummy region within one of the word line driver or the control circuit; arranging the plurality of memory macro layout diagrams in rows and columns in an IC layout diagram; arranging a plurality of through-silicon via (TSV) regions in the IC layout diagram by placing a TSV region of the plurality of TSV regions within the dummy region of each memory macro layout diagram of the plurality of memory macro layout diagrams; and storing the IC layout diagram comprising the plurality of memory macro layout diagrams and the plurality of TSV regions in a storage device.

16. The method of claim 15, wherein the arranging the plurality of TSV regions in the IC layout diagram comprises placing additional TSV regions of the plurality of TSV regions between the rows of memory macro layout diagrams of the plurality of memory macro layout diagrams.

17. The method of claim 16, wherein the arranging the plurality of memory macro layout diagrams in rows in the IC layout diagram comprises arranging the rows having a first pitch, and the placing the TSV region of the plurality of TSV regions within the dummy region of each memory macro layout diagram of the plurality of memory macro layout diagrams and the placing the additional TSV regions of the plurality of TSV regions between the rows of memory macro layout diagrams of the plurality of memory macro layout diagrams comprise arranging the plurality of TSV regions having a second pitch half as large as the first pitch.

18. The method of claim 15, wherein the arranging the plurality of TSV regions in the IC layout diagram comprises arranging the plurality of TSV regions based on a power distribution structure

of an IC package.

19. The method of claim 15, wherein the arranging the plurality of TSV regions in the IC layout diagram comprises arranging the plurality of TSV regions based on one or more design criteria of a logic die separate from the IC layout diagram.

20. The method of claim 15, wherein each of the first memory array and the second memory array of each memory macro layout diagram of the plurality of memory macro layout diagrams comprises an array of static random-access memory (SRAM) cells.
